

STEP 5

Automated testing

21 pytest tests that check the data, the model's behaviour, the API contract and the drift logic.

Test suite

File	What it checks
test_data.py	Real data passes validation · no leakage column is used as a feature · missing columns, unknown categories and non-positive quantities are rejected · engineered features are created correctly
test_model.py	Model passes the quality gate · beats the always-Low baseline · only predicts Low/Medium/High · handles missing values · behaviour test : ordering far more than demand never lowers the High-waste probability
test_api.py	/health returns ok · /predict returns a valid level and probabilities summing to 1 · each request is logged · batch endpoint works · optional fields can be omitted · invalid category, negative quantity and bad date return HTTP 422
test_drift.py	A normal batch shows no drift (no false alarms) · a heatwave batch is detected on Weather and Storage_Temperature

How to run

In a terminal (venv active), from the project folder:

```
pytest          # short output
pytest -v       # one line per test
```

Inside a Jupyter notebook cell, prefix it with ! and use the kernel's Python:

```
import os, sys
os.chdir(r"C:\Users\priya\MLOPS\food-waste-mlops")
!{sys.executable} -m pytest -v
```

Expected result

```
..... [100%]
21 passed in 10s
```

Watch out

Typing pytest alone in a notebook cell gives `NameError: name 'pytest' is not defined`. A cell runs Python code, and pytest is a terminal command. Use the ! form above.

Why it matters

ML tests go beyond 'does the code run'. Behaviour tests (more over-ordering should mean more waste risk) catch models that score well but have learned the wrong thing. The API tests act as a contract with whoever calls the service.